

Scratch Simulation Project:

1. Earth Orbiting the Sun

Project Overview

This project is an interactive **space simulation** created using **Scratch block-based programming**. The simulation visually represents the **Earth revolving around the Sun**, along with **blinking stars** to enhance the realism of outer space. It demonstrates basic astronomical concepts through animation and logical programming blocks.

Objective

The main objective of this project is to:

- Simulate the **orbital motion of the Earth around the Sun**
- Create a dynamic space environment using **blinking stars**
- Apply **Scratch block coding** to control motion, timing, and visual effects
- Present scientific concepts in an engaging and interactive way

Description of the Simulation

When the simulation starts:

- The **Sun** remains fixed at the center of the screen.
- The **Earth moves in a circular orbit** around the Sun, representing planetary revolution.
- The **stars blink continuously**, giving a lively and realistic space background.
- All animations run simultaneously using Scratch's event-driven and loop-based blocks.

This simulation helps in understanding how **orbital motion** works in a simplified and visual manner.

Tools & Technologies Used

- Scratch (Block-Based Programming)
- Motion blocks for orbital movement
- Control blocks for looping and timing

- Looks blocks for blinking star effects
- Event blocks to start the simulation

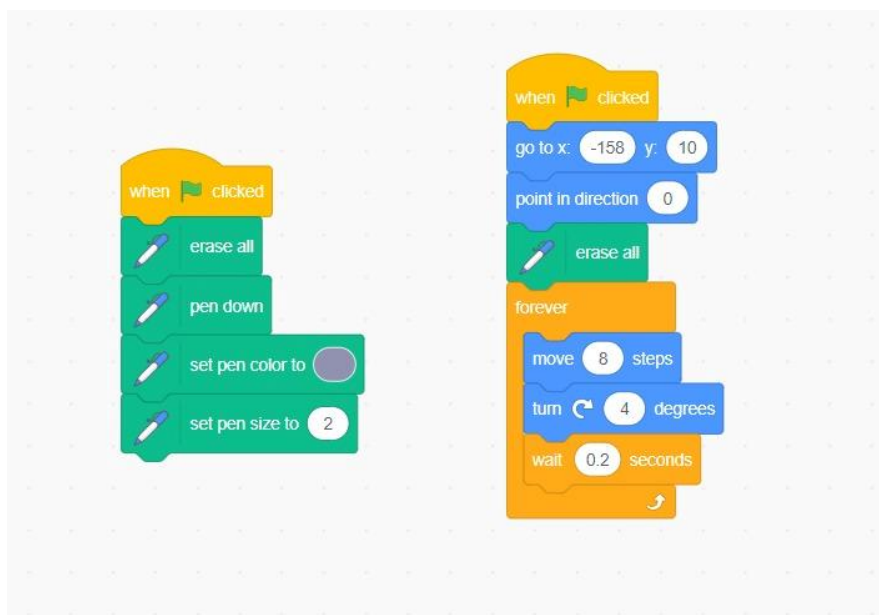
Key Features

- Circular orbital motion of Earth around the Sun
- Blinking stars effect for space realism
- Smooth animation using repeat and forever loops
- Beginner-friendly block code structure
- Educational and visually engaging simulation

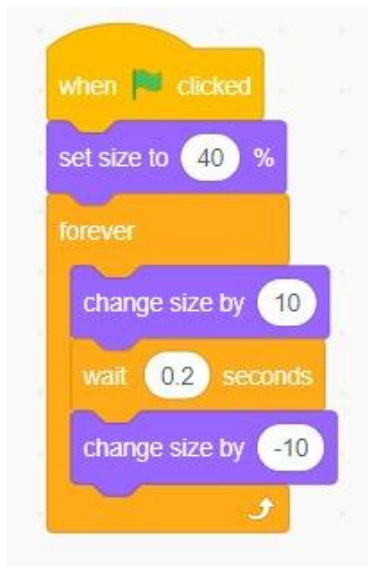
Code Implementation

Below are the Scratch block codes used to create the simulation:

📌 Earth Orbit Code Screenshot:



📌 Star Blinking Code Screenshot:



✦ Sun Setup Code Screenshot:



Learning Outcomes

Through this project, I gained:

- Hands-on experience with **Scratch animation and motion logic**
- Understanding of **orbital movement simulation**
- Skills in coordinating multiple sprites simultaneously
- Experience in converting scientific concepts into visual simulations

Conclusion

This Scratch-based simulation successfully demonstrates the **Earth's revolution around the Sun** using simple block programming.

2. Zombie Shooting Simulation

Project Overview

This project is an interactive **shooting game simulation** developed **using Scratch block-based programming**. The game features a **player character** who shoots bullets to eliminate **approaching zombies**. The simulation combines motion control, event handling, and collision detection to create a simple yet engaging gameplay experience.

Objective

The main objectives of this project are to:

- Design a **player-controlled shooting mechanism**
- Simulate **enemy (zombie) movement**
- Implement **bullet firing and collision logic**
- Develop an interactive game using Scratch blocks
- Improve logical thinking and game-design skills

Game Description

When the simulation starts:

- The **player sprite** appears on the screen and is ready to shoot.
- **Zombies move continuously** toward the player.
- The player can **shoot bullets** using control inputs.
- When a bullet touches a zombie, the **zombie is eliminated**.
- The game runs continuously using Scratch's event-driven and loop-based logic.

This simulation demonstrates the fundamentals of **2D shooting games** using block programming.

Sprites Used

- **Player** – Controlled by the user and responsible for shooting
- **Bullet** – Fired by the player to hit zombies
- **Zombie** – Enemy characters that move toward the player

Tools & Technologies Used

- **Scratch (Block-Based Programming)**
- Motion blocks for zombie movement
- Control blocks for looping and timing
- Sensing blocks for collision detection
- Event blocks for game start and shooting actions

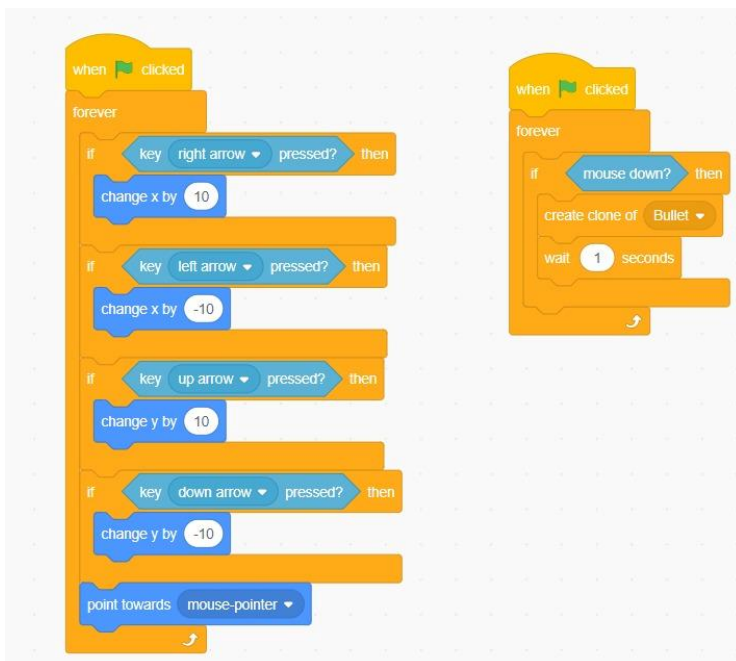
Key Features

- Player-controlled shooting mechanism
- Continuous zombie movement
- Bullet-zombie collision detection
- Simple and clean game logic
- Interactive and engaging gameplay

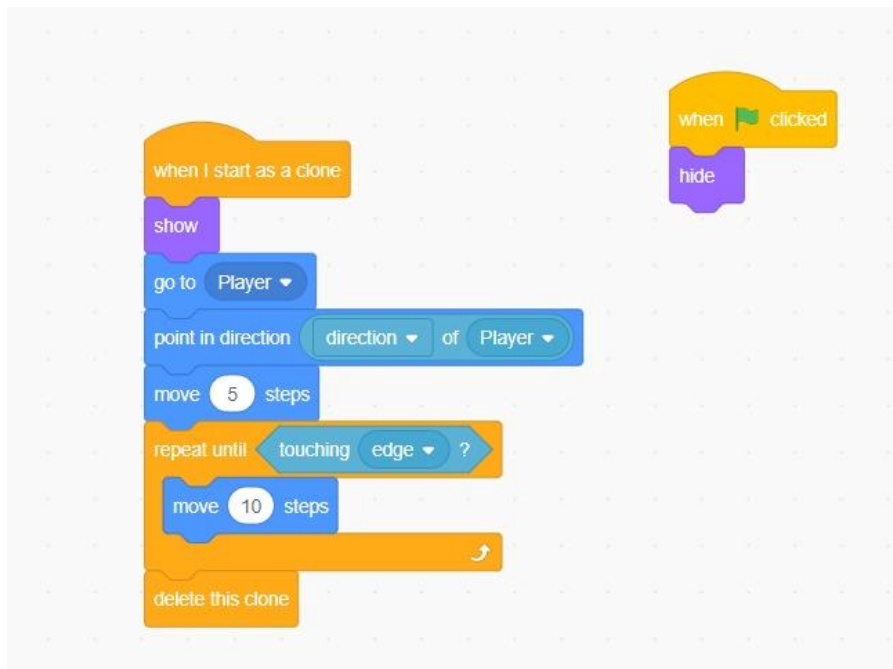
Code Implementation

The following Scratch block codes are used in this project:

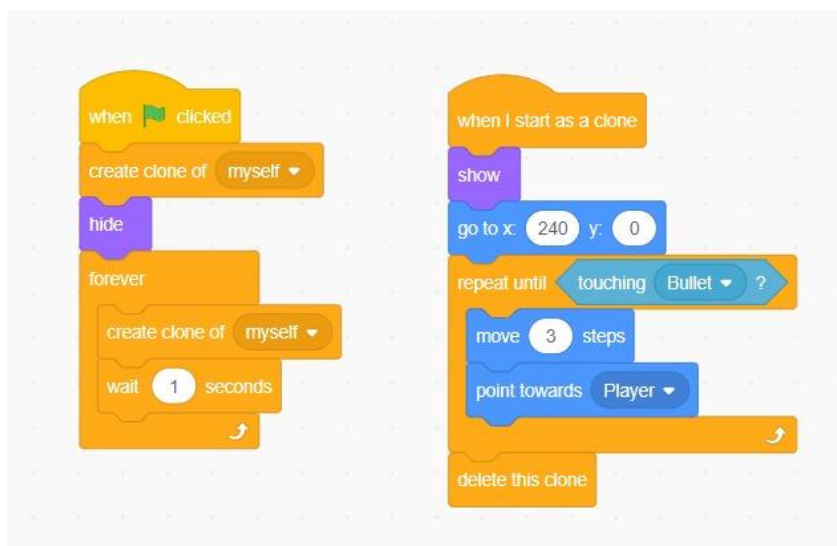
📌 Player Control & Shooting Code:



📌 Bullet Movement & Collision Code:



🚩 Zombie Movement Code:



Learning Outcomes

Through this project, I gained:

- Experience in game logic development using Scratch
- Understanding of event-based programming
- Knowledge of collision detection and sprite interaction
- Skills in designing simple 2D games using block code

Conclusion

This Scratch-based zombie shooting simulation successfully demonstrates how interactive games can be built using block programming.